

Step 1.

The screenshot shows the MySQL Workbench interface. The 'Calls' table has been created with the following schema:

```
CREATE TABLE calls(  
  ID CHAR(50),  
  cost_name CHAR(50),  
  sentiment CHAR(20),  
  csat_score INT,  
  call_timestamp CHAR(50),  
  reason CHAR(20),  
  city CHAR(20),  
  state CHAR(20),  
  channel CHAR(20),  
  response_time CHAR(20),  
  call_duration_minutes INT,  
  call_center CHAR(20)  
);
```

The 'Result Grid' shows the following data:

call_center	response_time	count
Baltimore/MD	Within SLA	6855
Baltimore/MD	Below SLA	2768
Baltimore/MD	Above SLA	1389
Chicago/IL	Within SLA	3361
Chicago/IL	Below SLA	1361
Chicago/IL	Above SLA	697
Denver/CO	Within SLA	1741
Denver/CO	Below SLA	692
Denver/CO	Above SLA	343
Los Angeles/CA	Within SLA	8668
Los Angeles/CA	Below SLA	3327
Los Angeles/CA	Above SLA	1739

The 'Output' pane shows the execution of a query that inserts data into the 'calls' table. The query is as follows:

```
SELECT DAYNAME(call_timestamp) AS Day_of_call, COUNT(*) num_of_calls FROM calls GROUP BY 1 ORDER BY 2 DESC LIMIT 0, 1000  
SELECT MIN(call_score) AS min_score, MAX(call_score) AS max_score, ROUND(AVG(call_score), 1) AS avg_score FROM calls WHERE csat_sc...  
SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls LIMIT 0, 1000  
SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call...  
SELECT call_center, response_time, COUNT(*) AS count FROM calls GROUP BY 1, 2 ORDER BY 3 DESC LIMIT 0, 1000  
SELECT call_center, response_time, COUNT(*) AS count FROM calls GROUP BY 1, 2 ORDER BY 3 DESC LIMIT 0, 1000
```

Step 2.

The screenshot shows the MySQL Workbench interface. The 'calls' table has been updated with the following data:

```
UPDATE calls SET call_timestamp = str_to_date(call_timestamp, '%m/%d/%Y');  
UPDATE calls SET csat_score = NULL WHERE csat_score = 0;  
SELECT * FROM calls LIMIT 10;
```

The 'Result Grid' shows the following data:

call_center	response_time	count
Baltimore/MD	Within SLA	6855
Baltimore/MD	Below SLA	2768
Baltimore/MD	Above SLA	1389
Chicago/IL	Within SLA	3361
Chicago/IL	Below SLA	1361
Chicago/IL	Above SLA	697
Denver/CO	Within SLA	1741
Denver/CO	Below SLA	692
Denver/CO	Above SLA	343
Los Angeles/CA	Within SLA	8668
Los Angeles/CA	Below SLA	3327
Los Angeles/CA	Above SLA	1739

The 'Output' pane shows the execution of a query that updates the 'calls' table. The query is as follows:

```
SELECT COUNT(*) AS rows_num FROM calls;  
SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls';  
SELECT DISTINCT sentiment FROM calls;  
SELECT DISTINCT reason FROM calls;
```

Step 3.

The screenshot shows the MySQL Workbench interface. The 'calls' table has been updated with the following data:

```
SELECT * FROM calls LIMIT 10;  
SELECT COUNT(*) AS rows_num FROM calls;  
SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls';  
SELECT DISTINCT sentiment FROM calls;
```

The 'Result Grid' shows the following data:

call_center	response_time	count
Baltimore/MD	Within SLA	6855
Baltimore/MD	Below SLA	2768
Baltimore/MD	Above SLA	1389
Chicago/IL	Within SLA	3361
Chicago/IL	Below SLA	1361
Chicago/IL	Above SLA	697
Denver/CO	Within SLA	1741
Denver/CO	Below SLA	692
Denver/CO	Above SLA	343
Los Angeles/CA	Within SLA	8668
Los Angeles/CA	Below SLA	3327
Los Angeles/CA	Above SLA	1739

The 'Output' pane shows the execution of a query that selects data from the 'calls' table. The query is as follows:

```
SELECT * FROM calls LIMIT 10;  
SELECT COUNT(*) AS rows_num FROM calls;  
SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls';  
SELECT DISTINCT sentiment FROM calls;
```

Step 4.

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Navigator

Schemas

- Filter objects
- phpmyadmin
- project
 - Tables
 - calls
 - Views
 - Stored Procedures
 - Functions
 - shipment_db
 - test

Query 1

```
19 SET SQL_SAFE_UPDATES = 1;
20 SELECT * FROM calls LIMIT 10;
21
22 SELECT COUNT(*) AS rows_num FROM calls;
23 SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls';
24
25 SELECT DISTINCT sentiment FROM calls;
26 SELECT DISTINCT reason FROM calls;
27 SELECT DISTINCT channel FROM calls;
28 SELECT DISTINCT response_time FROM calls;
29 SELECT DISTINCT call_center FROM calls;
30
31 SELECT sentiment, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
32 FROM calls GROUP BY 1 ORDER BY 3 DESC;
```

Result Grid

call_center
Los Angeles/CA
Baltimore/MD
Denver/CO
Chicago/IL

Schema: project

calls 43 calls 44 calls 45 calls 46 calls 47 x

Output

#	Time	Action	Message	Duration / Fetch
140	12:49:09	SELECT call_center, response_time, COUNT(*) AS count FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC LIMIT 0, 1000	12 rows returned	0.063 sec / 0.000 sec
141	12:51:37	SELECT DISTINCT sentiment FROM calls LIMIT 0, 1000	5 rows returned	0.047 sec / 0.000 sec
142	12:51:37	SELECT DISTINCT reason FROM calls LIMIT 0, 1000	3 rows returned	0.047 sec / 0.000 sec
143	12:51:37	SELECT DISTINCT channel FROM calls LIMIT 0, 1000	4 rows returned	0.031 sec / 0.000 sec
144	12:51:37	SELECT DISTINCT response_time FROM calls LIMIT 0, 1000	3 rows returned	0.031 sec / 0.000 sec
145	12:51:37	SELECT DISTINCT call_center FROM calls LIMIT 0, 1000	4 rows returned	0.031 sec / 0.000 sec

Object Info Session

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Step 5.

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Navigator

Schemas

- Filter objects
- phpmyadmin
- project
 - Tables
 - calls
 - Views
 - Stored Procedures
 - Functions
 - shipment_db
 - test

Query 1

```
28 SELECT DISTINCT response_time FROM calls;
29 SELECT DISTINCT call_center FROM calls;
30
31 SELECT sentiment, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
32 FROM calls GROUP BY 1 ORDER BY 3 DESC;
33
34 SELECT reason, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
35 FROM calls GROUP BY 1 ORDER BY 3 DESC;
36
37 SELECT channel, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
38 FROM calls GROUP BY 1 ORDER BY 3 DESC;
39
40 SELECT response_time, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
41 FROM calls GROUP BY 1 ORDER BY 3 DESC;
42
43 SELECT call_center, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
44 FROM calls GROUP BY 1 ORDER BY 3 DESC;
45
46 SELECT state, count(*)
47 FROM calls GROUP BY 1 ORDER BY 2 DESC;
48
49 SELECT DAYNAME(call_timestamp) as Day_of_call, COUNT(*) num_of_calls FROM calls GROUP BY 1 ORDER BY 2 DESC;
```

Result Grid

state	COUNT(*)
California	3631
Texas	3572
Florida	2834
New York	1786
Virginia	1164
Ohio	1160
District of Columbia	1110

Schema: project

Result 48 Result 49 Result 50 Result 51 Result 52 Result 53 x

Output

#	Time	Action	Message	Duration / Fetch
146	12:51:56	SELECT sentiment, count(*), ROUND(COUNT(*) / (SELECT COUNT(*) FROM calls) * 100, 1) AS pct FROM calls GROUP BY 1 ORDER BY 3 DESC	5 rows returned	0.031 sec / 0.000 sec
147	12:51:56	SELECT reason, count(*), ROUND(COUNT(*) / (SELECT COUNT(*) FROM calls) * 100, 1) AS pct FROM calls GROUP BY 1 ORDER BY 3 DESC	3 rows returned	0.047 sec / 0.000 sec
148	12:51:56	SELECT channel, count(*), ROUND(COUNT(*) / (SELECT COUNT(*) FROM calls) * 100, 1) AS pct FROM calls GROUP BY 1 ORDER BY 3 DESC	4 rows returned	0.062 sec / 0.000 sec
149	12:51:56	SELECT response_time, count(*), ROUND(COUNT(*) / (SELECT COUNT(*) FROM calls) * 100, 1) AS pct FROM calls GROUP BY 1 ORDER BY 3 DESC	3 rows returned	0.031 sec / 0.000 sec
150	12:51:56	SELECT call_center, count(*), ROUND(COUNT(*) / (SELECT COUNT(*) FROM calls) * 100, 1) AS pct FROM calls GROUP BY 1 ORDER BY 3 DESC	4 rows returned	0.031 sec / 0.000 sec
151	12:51:56	SELECT state, count(*) FROM calls GROUP BY 1 ORDER BY 2 DESC LIMIT 0, 1000	51 rows returned	0.047 sec / 0.000 sec

Object Info Session

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Step 6.

The screenshot shows the MySQL Workbench interface. The SQL editor contains a query that calculates various statistics from a 'calls' table, including counts, averages, and percentages for different reasons, channels, response times, call centers, states, and sentiments. The query is executed, and the results are displayed in a table with columns 'Day_of_call' and 'num_of_calls'.

```
34 * SELECT reason, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
35 FROM calls GROUP BY 1 ORDER BY 3 DESC;
36
37 * SELECT channel, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
38 FROM calls GROUP BY 1 ORDER BY 3 DESC;
39
40 * SELECT response_time, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
41 FROM calls GROUP BY 1 ORDER BY 3 DESC;
42
43 * SELECT call_center, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
44 FROM calls GROUP BY 1 ORDER BY 3 DESC;
45
46 * SELECT state, count(*)
47 FROM calls GROUP BY 1 ORDER BY 2 DESC;
48
49 * SELECT DAYNAME(call_timestamp) AS Day_of_call, COUNT(*) num_of_calls FROM calls GROUP BY 1 ORDER BY 2 DESC;
50
51 * SELECT MIN(csat_score) AS min_score, MAX(csat_score) AS max_score, ROUND (AVG(csat_score), 1) AS avg_score
52 FROM calls WHERE csat_score != 0; # MySQL added to blank rows. But the min is 1.
53
54 * SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls;
```

Day_of_call	num_of_calls
Friday	5570
Thursday	5481
Wednesday	4449
Tuesday	4408
Saturday	4403
Monday	4334
Sunday	4296

Step 7.

The screenshot shows the MySQL Workbench interface. The SQL editor contains a query that calculates various statistics from a 'calls' table, including counts, averages, and percentages for different reasons, channels, response times, call centers, states, and sentiments. The query is executed, and the results are displayed in a table with columns 'min_call_duration', 'max_call_duration', and 'avg_call_duration'.

```
51 * SELECT MIN(csat_score) AS min_score, MAX(csat_score) AS max_score, ROUND (AVG(csat_score), 1) AS avg_score
52 FROM calls WHERE csat_score != 0; # MySQL added to blank rows. But the min is 1.
53
54 * SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls;
55
56 * SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call_duration FROM calls;
57
58 * SELECT call_center, response time, COUNT(*) AS count
59 FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
60
61 * SELECT call_center, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
62
63 * SELECT channel, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
64
65 * SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;
66
67 * SELECT state, reason, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;
68
69 * SELECT state, sentiment, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;
70
71 * SELECT state, AVG(csat_score) AS avg_csat_score FROM calls WHERE csat_score != 0 GROUP BY 1 ORDER BY 2 DESC;
```

min_call_duration	max_call_duration	avg_call_duration
15	45	23.0212

Step 8.

MySQL Workbench

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Navigator

Schemas

Filter objects

phmyadmin

project

Tables

calls

Views

Stored Procedures

Functions

shipment_db

test

Query 1

```

53 FROM calls WHERE csat_score != 0; # MySQL added to blank rows. But the min is 1.
54
55 SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls;
56
57 SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call_duration FROM calls
58
59 SELECT call_center, response_time, COUNT(*) AS count
60
61 FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
62
63 SELECT call_center, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
64
65 SELECT channel, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
66
67 SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;
68
69 SELECT state, reason, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;
70
71 SELECT state, sentiment, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
72
73 SELECT state, AVG(csat_score) AS avg_csat_score FROM calls WHERE csat_score != 0 GROUP BY 1 ORDER BY 2 DESC;
74
75 SELECT sentiment, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
76

```

Result Grid

sentiment	AVG(call_duration_minutes)
Negative	25.2618
Neutral	24.9398
Very Negative	24.9391
Positive	24.8620
Very Positive	24.7593

Schema: project

Object Info Session

Output

#	Time	Action	Message	Duration / Fetch
152	12:52:12	SELECT DAYNAME(call_timestamp) AS Day_of_call, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC LIMIT 0, 1000	7 rows returned	0.047 sec / 0.000 sec
153	12:52:55	SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls LIMIT 0, 1000	1 row(s) returned	0.031 sec / 0.000 sec
154	12:52:55	SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call...	1 row(s) returned	0.005 sec / 0.000 sec
155	12:52:55	SELECT call_center, response_time, COUNT(*) AS count FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC LIMIT 0, 1000	Error Code: 1054. Unknown column 'response' in field list	0.000 sec
156	12:53:09	SELECT state, AVG(csat_score) AS avg_csat_score FROM calls WHERE csat_score != 0 GROUP BY 1 ORDER BY 2 DESC LIMIT 0, 1000	51 rows returned	0.032 sec / 0.000 sec
157	12:53:09	SELECT sentiment, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC LIMIT 0, 1000	5 rows returned	0.031 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Step 9.

MySQL Workbench

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File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

Filter objects

phmyadmin

project

Tables

calls

Views

Stored Procedures

Functions

shipment_db

test

Query 1

```

56
57 SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call_duration FROM calls
58
59 SELECT call_center, response_time, COUNT(*) AS count
60
61 FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
62
63 SELECT call_center, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
64
65 SELECT channel, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
66
67 SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;
68
69 SELECT state, reason, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;
70
71 SELECT state, sentiment, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
72
73 SELECT state, AVG(csat_score) AS avg_csat_score FROM calls WHERE csat_score != 0 GROUP BY 1 ORDER BY 2 DESC;
74
75 SELECT sentiment, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
76
77
78 SELECT call_center, response_time, COUNT(*) AS count FROM calls group by 1,2 ORDER BY 1,3 DESC;
79

```

Result Grid

call_center	response_time	count
BaltimoreMD	Within SLA	6855
BaltimoreMD	Below SLA	2768
BaltimoreMD	Above SLA	1389
ChicagoIL	Within SLA	3061
ChicagoIL	Below SLA	1361

Schema: project

Object Info Session

Output

#	Time	Action	Message	Duration / Fetch
153	12:52:55	SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls LIMIT 0, 1000	1 row(s) returned	0.031 sec / 0.000 sec
154	12:52:55	SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call...	1 row(s) returned	0.000 sec / 0.000 sec
155	12:52:55	SELECT call_center, response_time, COUNT(*) AS count FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC LIMIT 0, 1000	Error Code: 1054. Unknown column 'response' in field list	0.000 sec
156	12:53:09	SELECT state, AVG(csat_score) AS avg_csat_score FROM calls WHERE csat_score != 0 GROUP BY 1 ORDER BY 2 DESC LIMIT 0, 1000	51 rows returned	0.032 sec / 0.000 sec
157	12:53:09	SELECT sentiment, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC LIMIT 0, 1000	5 rows returned	0.031 sec / 0.000 sec
158	12:53:23	SELECT call_center, response_time, COUNT(*) AS count FROM calls group by 1,2 ORDER BY 1,3 DESC LIMIT 0, 1000	12 rows returned	0.062 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.